

Validation-Selected Hybrid Occupancy Forecasting for Empty-Window Recommendations: An Offline Post-Bin Case Study

Junyu Chen

The Storm King School
Cornwall-on-Hudson, New York, USA
Email: stevenchen0326@hotmail.com

Chong Li

Department of Electrical Engineering
Columbia University
New York, NY, USA
cl3607@columbia.edu

Abstract—Building operations need occupancy forecasts that arrive in time, last long enough, and rarely conflict with later occupancy observations. We study this problem in a single-building, 15-min, offline *post-bin* case study. The primary *hybrid* is a weighted average of Empty-class scores from a historical schedule, LightGBM, and a compact Transformer, combining recurring, nonlinear, and sequential patterns. Validation area under the precision–recall curve (AUPRC) selects the weights. AUPRC summarizes Empty-class precision and recall across score thresholds; higher values indicate better discrimination when class frequencies are unequal. A separate validation search selects the operating threshold that maximizes a load-opportunity proxy under a 10% conflict-rate limit and a one-hour minimum window. Conflict rate is the share of recommended intervals later labelled occupied. The opportunity proxy sums processed HVAC-plus-lighting load in recommendations later labelled Empty and represents load available for further operational investigation. A stored anchor t covers the completed input bin $[t, t + 15 \text{ min})$, making the forecast available at $t + 15 \text{ min}$. Across 388,032 overlapping held-out forecast rows, hybrid AUPRC is 0.8514, compared with 0.8497 for the historical baseline and 0.8382 for LightGBM. Across 43 non-overlapping policy horizons, the fixed hybrid policy recommends 259 intervals; all are subsequently camera-label-empty, form 14 label-safe windows, and coincide with 490.1 kWh of processed load proxy. LightGBM finds a similar opportunity (493.9 kWh) with 11 conflicts, whereas the conflict-free historical policy finds 94.6 kWh. The hybrid therefore preserves strong forecast discrimination while improving the observed opportunity–conflict trade-off. A prospective rerun with verified source timestamps is the next step toward operational evaluation.

Index Terms—occupancy forecasting, hybrid forecasting, building energy management, offline load opportunity, post-bin evaluation, stable empty windows

I. INTRODUCTION

Building energy-flexibility research considers how operational demand can be adjusted while respecting service constraints [1], [2]. Occupancy is an important contextual signal for that task: it can identify intervals in which a building zone may be empty and, therefore, potentially suitable for a carefully constrained operational action. Occupancy modeling has long been studied for building energy management [3], and recent reviews show a broad forecasting literature spanning data acquisition, estimation, prediction, and evaluation [4]–

[6]. Operational recommendations add three requirements to forecast discrimination: timely issuance, useful duration, and an acceptable observed conflict rate.

In plain terms, the analysis has two stages. First, three models estimate how strongly each future 15-min interval resembles an Empty interval. Their scores are combined in a *hybrid*: a weighted average of model outputs. Second, a decision rule converts sufficiently high scores into recommendations that continue for at least one hour. Conflict rate measures the share of recommended intervals later labelled occupied by the cameras. A load-opportunity proxy records processed HVAC-plus-lighting load during recommended intervals later labelled Empty.

Occupancy-prediction-based cooling-control studies and HVAC-control evaluations assess modeled or field operating interventions [7]–[9]. This study measures an *offline safe load opportunity*: the processed HVAC-plus-lighting proxy in intervals recommended by a fixed rule and subsequently labelled Empty by the cameras. The statistic screens candidate intervals for operational study. Energy-savings estimation requires a control action, a counterfactual, and explicit comfort and equipment constraints.

The repository’s left-labelled records determine forecast timing. A row labelled 00:00 summarizes $[00:00, 00:15)$ and becomes available at 00:15, so the study evaluates post-bin next-record forecasts. Section VI describes the source-timestamp and imputation provenance available in the cleaned release.

The research question is: *for one office-building time series, how do validation-selected Empty-class scores trade off forecast discrimination, subsequently observed camera-label conflict, and offline safe load opportunity when recommendations must form at least one-hour stable windows?* Forecast discrimination measures how well the models rank future Empty intervals across all overlapping rolling 24-h forecasts. Policy evaluation converts those scores into recommendations on non-overlapping midnight-labelled daily horizons, counting each target interval once.

The paper makes three contributions. First, it documents a post-bin chronological protocol that maps 24-h Empty-

class scores to stable-window recommendations and reserves controllable-load data for downstream evaluation. Second, it evaluates a validation-selected hybrid of a historical schedule baseline, LightGBM, and a compact Transformer. The weights optimize validation forecast discrimination; the operating threshold maximizes the validation load-opportunity proxy under a 10% empirical conflict-rate limit. Third, it reports the uncertainty behind the point estimates. The hybrid has slightly higher AUPRC than the two baselines and zero observed camera-label conflicts in the held-out policy period. Paired test contrasts cross zero, and validation perturbations reveal several competitive weights and thresholds.

Section II positions the evaluation against occupancy forecasting and control research. Sections III and IV define the data, timing, policy, and experimental protocol. Section V reports the canonical results, and Sections VI–VII discuss their limits and implications. Appendix A records the status of the expanded decision-aware and window-aware searches.

II. RELATED WORK

A. Occupancy Forecasting

Occupancy forecasting is a mature building-analytics problem. Reviews by Jin *et al.* and Caballero-Peña *et al.* organize the literature around forecasting applications, data sources, and evaluation processes [4], [5]. Earlier commercial-building comparisons include stochastic and machine-learning approaches [10]. Work based on indoor environmental measurements illustrates the potential and data-quality dependence of sensor-assisted prediction [11]. Sequence models also remain active in the field, including Transformer-based occupancy prediction [12]. Together, these studies motivate comparing schedule, tabular, and sequence models under a common operating rule.

The present study fits a schedule prior on the training partition and compares it with nonlinear tabular and sequence models. The tabular reference uses LightGBM [13], the random-forest comparator follows Breiman [14], and the sequence comparator is a compact encoder-only Transformer with a known-future calendar projection, architecturally motivated by [15]. The saved artifact identifier `DLinear` denotes a direct one-layer occupancy-history baseline. The decomposition-based `DLinear` architecture of Zeng *et al.* uses a different structure [16]. The comparison tests whether combining the established components produces an inspectable, validation-selected offline policy under the same target and time split.

B. From Occupancy Prediction to Building Operation

Several studies couple occupancy predictions to a downstream HVAC or ventilation decision. Peng *et al.* evaluate occupancy-prediction-based cooling control [7]. Esrafilian-Najafabadi and Haghighat analyze how occupancy-model choices relate to HVAC control performance [8], and Zhuang *et al.* develop probabilistic occupancy forecasts for risk-aware ventilation [17]. Wietzke *et al.* further connect occupancy prediction to model predictive control and simulation [18].

These studies show why forecast discrimination and downstream trade-offs deserve separate evaluation.

Lin *et al.* report an occupancy-based HVAC control-system evaluation in an office building [9]. The present retrospective study uses *opportunity* for a processed meter-derived load proxy that coincides with a recommended and camera-label-empty interval. An intervention or simulation is the appropriate setting for subsequent savings and control-performance evaluation.

C. Decision-Layer Evaluation

Decision-focused learning and predict-then-optimize methods train predictive models against a downstream decision objective [19], [20]. Here, candidate score models are fixed, blend weights are selected by validation AUPRC, and a subsequent score-threshold search uses an empirical validation conflict cutoff and offline opportunity objective. We therefore describe the pipeline as a *validation-selected decision-layer evaluation*.

III. DATA, FORECASTING, AND RECOMMENDATION METHOD

A. Data and Information Available at Prediction Time

The source is the cleaned LBNL Building 59 release described by Luo *et al.* [21] and distributed through Dryad [22]. The repository retains one selected south-zone subset and resamples the imported streams to 15-min bins. An interval receives the camera-derived occupied label when the maximum selected south-zone camera count is greater than zero. The analysis flips this label so that $y_{d,t} = 1$ denotes EMPTY and $y_{d,t} = 0$ denotes camera-labeled occupied. The dataset therefore operationalizes EMPTY as zero detected occupants in the selected camera counts. Independent physical-presence validation is a future data-collection step. The analysis contains 26,413 local-time rows from May 2018 through February 2019. Table I gives the exact split and policy scopes.

Naive source timestamps are interpreted as UTC and converted to `America/Los_Angeles` before time features are formed. A solar-radiation peak sanity check supports this interpretation; the source documentation leaves the timestamp time-zone unspecified. Each resampled record uses a left-closed, left-labelled convention: its label t denotes $[t, t+15 \text{ min})$. The tabular and sequence models consume the completed anchor record, giving an effective availability boundary of $t+15 \text{ min}$. The first target bin begins at that boundary, and a 96-step horizon spans $[t+15 \text{ min}, t+24 \text{ h}+15 \text{ min})$. A policy anchor labelled 00:00 therefore becomes available at 00:15 and targets the following 96 bins.

Calendar variables are known for future target times. The tabular information set comprises past occupancy/count lags and rolling summaries, completed-input-bin Wi-Fi, temperature, and weather values, and target-time calendar variables. After import, environmental and association-count sensors are forward-filled in row order; leading missing values are set to zero. Repository-level imputation provenance begins after import of the cleaned release. Wi-Fi has 81.57% missingness

TABLE I
DATA AND CANONICAL EVALUATION PROTOCOL.

Item	Canonical setting
Data scope	Cleaned Building 59 selected south-zone streams; 26,413 15-min local-time rows, 22 May 2018–21 Feb. 2019 02:00
Empty target	Camera-derived Empty = 1 iff the maximum selected south-zone count is zero in the 15-min bin
Chronological splits	Train: 23 May–25 Nov. 2018 (32.65% Empty); validation: 28 Nov. 2018–6 Jan. 2019 (29.64%); test: 9 Jan.–21 Feb. 2019 (27.75%)
Boundary separation	24.25-h train–validation and validation–test gaps
Input-bin timing	Anchor label t denotes completed left-labelled $[t, t + 15 \text{ min})$ bin; effective boundary $t + 15 \text{ min}$
Forecast scope	96 history and 96 target steps; targets $[t + 15 \text{ min}, t + 24 \text{ h} + 15 \text{ min})$; all overlapping test forecasts
Policy scope	39 validation and 43 test midnight-labelled, non-overlapping horizons; effective test boundary 00:15
Stable recommendation	Contiguous Empty score at or above threshold for ≥ 4 steps (1 h), extracted independently in each policy horizon
Opportunity proxy	Post-import row-order-filled $\max(\text{HVAC}_S + \text{lighting}_S, 0) \times 0.25 \text{ h}$ restricted to recommended-and-camera-label-empty intervals
Source limit	Prospective rerun requires original stream timestamps and upstream imputation lineage

before the unlimited row-order forward fill. The post-forecast opportunity calculation uses the retained HVAC and lighting series; other electrical-load streams and their aggregate remain outside the analysis.

Each rolling forecast uses 96 historical 15-min intervals and predicts the next 96 intervals. Chronological train, validation, and test segments are separated by 24.25-h gaps, keeping every saved forecast history and target window within its assigned split. Fitting, policy selection, and held-out saved-output evaluation therefore use distinct temporal segments.

B. Forecast Models and Why Use a Hybrid

The comparison includes a historical weekday/time-slot average fitted on the training partition, LightGBM, random forest, a direct linear occupancy baseline (legacy artifact key `DLinear`), and a compact Transformer encoder (legacy artifact key `Original Transformer`). Table II gives the stored implementation settings. The saved outputs are bounded $[0, 1]$ Empty-class scores, denoted by p . The pipeline stores them as raw, uncalibrated scores. Historical runs bear seed labels 42, 43, and 44, and their stored scores are pointwise averaged. Deep modules were instantiated before the seed-reset function, so controlled seed-dispersion analysis requires corrected retraining.

The hybrid combines three sources of predictive structure. The historical model represents the building’s recurring week-day and time-of-day schedule. LightGBM represents nonlin-

ear relationships among lagged occupancy, sensor, weather, and calendar features. The compact Transformer represents patterns across the recent 96-interval sequence. A single validation step combines the stored component outputs.

The saved primary hybrid blends the three component Empty-class scores. For time t it is

$$p_H(t) = 0.15p_{\text{hist}}(t) + 0.60p_{\text{LGBM}}(t) + 0.25p_{\text{Tr}}(t), \quad (1)$$

where the nonnegative weights sum to one. These values attain the maximum validation AUPRC on a declared 0.05 simplex grid of 231 legal weights. The pipeline loads test scores after this canonical selection and preserves the fitted component models. Section V-B evaluates the small separation among validation candidates.

Forecast selection and recommendation selection answer different questions. The weight search uses validation AUPRC to choose a score that ranks Empty intervals. After those weights are fixed, the threshold search described next chooses when to act by balancing two policy quantities: the observed camera-label conflict rate and the offline load-opportunity proxy. Both selection steps use validation data, and the pipeline fixes their outputs before reading test conflict and opportunity outcomes.

C. From Forecast Scores to One-Hour Recommendations

For a model-specific score threshold τ , define an initial recommendation $b_{d,t} = \mathbf{1}\{p_{d,t} \geq \tau\}$. Let $L = 4$ be the number of 15-min intervals in one hour. The policy recommendation $r_{d,t}$ equals one when $b_{d,t}$ belongs to a contiguous run of at least L intervals within a midnight-labelled daily horizon. Runs are extracted independently within each 24-h forecast horizon. The interval camera-label-conflict rate is

$$C = \frac{\sum_{d,t} r_{d,t}(1 - y_{d,t})}{\sum_{d,t} r_{d,t}}, \quad (2)$$

For an empty recommendation set, define $C = 0$.

Let $\tilde{h}_{d,t}$ and $\tilde{\ell}_{d,t}$ denote the processed south-HVAC and south-lighting meter streams, respectively, in nominal kW. Define

$$q_{d,t} = 0.25 \max(\tilde{h}_{d,t} + \tilde{\ell}_{d,t}, 0), \quad (3)$$

which is a processed meter-derived 15-min load proxy in kWh. The nominal full-metered accounting convention uses a coefficient of one for each stream, and multiplication by 0.25 h converts kW to interval kWh. Each stream is 8.03% missing before the post-import row-order forward fill, so q includes carried-forward values; its momentary controllability is untested. The paper reports

$$O_{\text{safe}} = \sum_{d,t} r_{d,t} y_{d,t} q_{d,t}. \quad (4)$$

Equation (4) is an offline overlap measure: it sums the processed load proxy where a recommendation and a subsequently observed camera-derived Empty label coincide. Estimating the effect of a control action would require a counterfactual. For each model, τ is selected from

TABLE II

STORED MODEL SPECIFICATIONS AND INFORMATION AVAILABILITY. SAVED NONDETERMINISTIC OUTPUTS AVERAGE HISTORICAL RUNS LABELLED 42, 43, AND 44; DEEP-MODULE CONSTRUCTION PRECEDED SEED RESET, LIMITING SEED-DISPERSION ANALYSIS.

Model	Inputs and training set	Stored implementation setting
Historical average	Known target weekday and 15-min slot; training partition	Deterministic mean occupied label by weekday/time slot, with the train mean as fallback
LightGBM and random forest	Occupancy/count lags and rolling summaries; completed-input-bin sensor values; known-future calendar features; at most 200,000 stratified training rows per seed	LightGBM: 320 trees, learning rate 0.05, 31 leaves, minimum child 50, column fraction 0.85, full-row sampling (<code>subsample=1</code> , <code>subsample_freq=0</code>). Random forest: 70 trees, depth 14, minimum leaf 20, square-root feature sampling
Direct linear occupancy base-line (legacy DLinear)	Past 96 occupied indicators	One direct 96-to-96 linear mapping on the raw occupancy history; maximum 12 epochs, validation-loss early stopping with patience 4
Compact encoder (legacy Transformer)	Transformer Original Past 96 occupancy/count, completed-input-bin sensor, and calendar values; known-future calendar values	One encoder layer, 32-dimensional state, 4 heads, 0.1 dropout; known-future calendar projection; AdamW at 10^{-3} ; maximum 12 epochs and validation-loss early stopping with patience 4

$\{0.050, 0.075, \dots, 0.950\}$ on validation midnight-labelled horizons to maximize O_{safe} subject to the empirical condition $C_{\text{val}} \leq 0.10$, breaking exact opportunity ties by Empty recall. Because O_{safe} uses realized labels and processed loads, it serves as an offline validation-selection objective. The selected threshold is then fixed for all test policy calculations.

IV. EXPERIMENTAL DESIGN

A. Two Evaluation Scopes and Their Metrics

Forecast metrics use all overlapping rolling test forecasts. Precision is the fraction of intervals predicted Empty that are later labelled Empty, while recall is the fraction of camera-labelled Empty intervals recovered by the model. Changing the score threshold trades precision against recall. The primary metric, Empty-class area under the precision–recall curve (AUPRC), summarizes that trade-off across thresholds; larger values indicate better discrimination. AUPRC is informative to report alongside other measures when the two classes have unequal frequencies [23]. Empty precision, recall, and F1 in the model-comparison table use a descriptive $p_{\text{Empty}} \geq 0.5$ cutoff. The operating policy uses the separately selected threshold in Eq. (2).

Policy metrics use a different, non-overlapping scope: 39 validation and 43 test midnight-labelled 96-step horizons. The latter have an effective 00:15 issue boundary under the completed-bin convention in Section III. For every model, the reported threshold comes from the validation policy horizons. The fixed test policy reports recommended, camera-label-safe, and conflict intervals; stable recommended and camera-label-safe windows; coverage; and Eq. (4). The temporal split and fixed-policy evaluation preserve time order when estimating time-series performance [24].

B. Uncertainty, Diagnostics, and Reproducibility

Uncertainty is quantified with 2,000 paired daily-block bootstrap resamples of the 43 test policy horizons while holding model weights and thresholds fixed. The resulting intervals are conditional on the selected model and policy; selection

uncertainty lies outside them. Brier score, log loss, and 10-bin quantile expected calibration error are reported as held-out diagnostics of the uncalibrated scores. A sensitivity analysis varies the minimum window length from 15 min to 4 h while preserving the validation-selected threshold.

The saved-output path validates aligned prediction exports, repeats the validation-only blend and threshold selection, and regenerates the canonical result artifacts from stored scores. End-to-end model retraining follows a separate workflow. Repository tests check split separation, timestamp offsets, convex-weight selection, policy accounting, and the requirement that the two exploratory selection routines freeze candidates before reading their test diagnostics.

C. Primary and Exploratory Analyses

The canonical analysis is the validation-selected primary hybrid in Eq. (1), with the same protocol applied to fixed reference models. The repository also contains a test-ranked balanced hybrid, an expanded joint weight-threshold search, and a window-aware search. Appendix A classifies these enlarged-search candidates as exploratory because their historical test diagnostics come from a previously inspected period.

V. RESULTS

A. Forecast Quality on Overlapping Test Forecasts

Table III reports the three canonical models central to the paper. The primary hybrid has the highest all-overlap Empty AUPRC point estimate (0.8514), followed closely by the historical average (0.8497); LightGBM reaches 0.8382. The hybrid also has the lowest Brier score among these three models (0.0928). These forecast results cover 388,032 overlapping rows from 4,042 anchors. Appendix A retains the extended seven-model comparison and explains the supplementary status of the test-ranked balanced hybrid.

The strong schedule baseline is a substantive result: weekday and time-of-day structure carries much of the Empty-class predictive signal in this building. The hybrid’s point AUPRC margins are 0.0017 over the historical baseline and 0.0132 over

TABLE III

CANONICAL THREE-MODEL FORECAST COMPARISON ON ALL OVERLAPPING HELD-OUT TEST FORECASTS (EMPTY = 1). SCORES ARE UNCALIBRATED. PRECISION, RECALL, AND F1 USE A DESCRIPTIVE 0.5 SCORE CUTOFF; TABLE V USES SEPARATELY SELECTED VALIDATION POLICY THRESHOLDS.

Model	Empty AUPRC	Empty Brier	Empty precision	Empty recall	Empty F1
Historical average	0.8497	0.0974	0.7264	0.8015	0.7621
LightGBM	0.8382	0.0956	0.7610	0.7593	0.7601
Primary hybrid	0.8514	0.0928	0.7677	0.7556	0.7616

LightGBM. Section V-C evaluates the corresponding daily-horizon differences.

B. Validation-Selection Stability

The canonical weight vector attains the nominal grid maximum of 0.728638 validation AUPRC. The runner-up is 0.000032 lower, and eight of the 231 legal vectors lie within 0.0005. Table IV reports separate validation-only perturbation diagnostics on 39 non-overlapping validation policy horizons. Nearby blends change rank under validation-horizon perturbation, and threshold selection varies across daily-block bootstrap resamples. The canonical vector is therefore one validation-selected solution within a competitive neighborhood.

C. Validation-Selected Conflict and Load-Opportunity Outcomes

Table V and Fig. 1 report the policy scope: 43 non-overlapping test horizons and 4,128 15-min target intervals. The horizons are anchored by midnight-labelled completed input bins and have an effective 00:15 availability boundary. Every listed threshold met the empirical validation 10% interval-conflict condition. The primary hybrid used $\tau = 0.875$ after a validation conflict rate of 8.75%. On test, it recommended 259 intervals. All 259 were subsequently camera-label-empty. Those intervals formed 14 recommended windows, all camera-label-safe under the observed label, and coincided with 490.1 kWh of offline camera-label-empty load-proxy overlap (11.40 kWh/horizon-day).

LightGBM produced a nearly equal 493.9 kWh opportunity, with 11 of 265 recommended intervals camera-labelled occupied (4.15%) and three of 19 recommended windows containing an occupancy conflict. The historical average was also conflict-free under this label and coincided with 94.6 kWh of processed load-proxy opportunity. Figure 1 compares these three pre-specified models.

Taken together, the forecast and policy results explain why the hybrid is the primary case-study model. It preserves the strong schedule baseline’s Empty-class discrimination while producing the highest nominal AUPRC of the three models. At its validation-selected threshold, it identifies nearly the same held-out load-opportunity proxy as LightGBM (490.1 versus 493.9 kWh) with fewer observed camera-label conflicts (0 versus 11), and substantially more opportunity than the conflict-free historical policy (490.1 versus 94.6 kWh). These results give the hybrid the strongest observed opportunity-conflict trade-off in the saved test period.

D. Conditional Uncertainty and Diagnostics

The daily-block uncertainty results in Table VI condition on the previously selected weights and thresholds. The primary hybrid’s daily-horizon AUPRC is 0.8522 with a 95% percentile interval of [0.7918, 0.9016]. Its aggregation unit differs from the all-overlap value in Table III. The paired AUPRC intervals for primary minus LightGBM and primary minus historical average both cross zero. The paired safe-opportunity interval crosses zero for primary minus LightGBM; the primary-minus-historical interval is positive. Figure 2 displays these paired contrasts on zero-centered axes.

The primary hybrid’s held-out score diagnostics are Brier score 0.0928, log loss 0.2984, and 10-bin quantile ECE 0.0253. At the fixed primary threshold, a one-hour or two-hour minimum window retains 490.1 kWh and zero observed camera-label conflicts; a four-hour requirement retains 407.8 kWh and 11 fully camera-label-safe windows. These sensitivity results preserve the selected threshold. Every observed primary-hybrid camera-label conflict block is zero, so a nonparametric bootstrap also returns a zero conflict interval. Under an independent-trial approximation, zero events yield one-sided 95% upper bounds of 1.15% for 259 intervals and 19.3% for 14 windows. The clustering of intervals and windows makes these bounds descriptive.

VI. DISCUSSION

A. Interpretation

The historical average’s performance shows that a stable schedule can provide a highly competitive Empty forecast in a single office building. The hybrid’s small point AUPRC advantage falls within the paired daily-block uncertainty intervals, so its value rests on the combined trade-off. In the saved test period, its selected operating point has approximately the same processed load-proxy opportunity as LightGBM and zero camera-labelled occupied recommendations. Forecast quality, conflict rate, and opportunity together provide a more informative comparison than forecast rank alone.

The offline opportunity formulation makes the recommendation rule, duration requirement, conflict denominator, and load proxy inspectable. The stable-window rule removes short isolated intervals. Equation (4) records the processed load coinciding with the remaining recommendations. Translating that overlap into operational benefit requires a counterfactual and constraints on equipment, indoor conditions, tariffs, occupants, and later load effects. Those additions would allow future work

TABLE IV

VALIDATION-ONLY SELECTION-STABILITY DIAGNOSTICS. THE BLEND PERTURBATION USES 39 NON-OVERLAPPING VALIDATION POLICY HORIZONS AND THE TOP 12 FULL-OVERLAP GRID CANDIDATES; THE CANONICAL ALL-OVERLAP SELECTION REMAINS PRIMARY. THRESHOLD DIAGNOSTICS APPLY THE DECLARED VALIDATION POLICY RULE.

Diagnostic	Result	Frequency / value
Full-overlap 231-weight grid	Nominal rank-1 AUPRC; rank-1 minus rank-2; candidates within 0.0005	0.728638; 0.000032; 8
Daily-block bootstrap, top-12 blends (1,000)	Canonical (0.15, 0.60, 0.25) selected; most frequent (0.05, 0.80, 0.15)	23 (2.3%); 365 (36.5%)
Leave-one-validation-horizon-out, blends (39)	top-12 Canonical selected; most frequent (0.05, 0.75, 0.20)	1 (2.6%); 31 (79.5%)
Daily-block bootstrap, primary threshold (1,000)	Canonical $\tau = 0.875$; $\tau = 0.850$; $\tau = 0.950$	111 (11.1%); 376 (37.6%); 375 (37.5%)
Leave-one-validation-horizon-out, primary threshold (39)	Canonical $\tau = 0.875$	36 (92.3%)

TABLE V

VALIDATION-SELECTED ONE-HOUR STABLE-WINDOW POLICIES EVALUATED ON 43 NON-OVERLAPPING TEST HORIZONS. ANCHORS ARE MIDNIGHT-LABELLED COMPLETED INPUT BINS WITH EFFECTIVE 00:15 BOUNDARIES. EVERY THRESHOLD MAXIMIZES VALIDATION SAFE OPPORTUNITY SUBJECT TO EMPIRICAL VALIDATION INTERVAL CONFLICT $\leq 10\%$. “SAFE” MEANS RECOMMENDED AND SUBSEQUENTLY CAMERA-LABEL-EMPTY; OPPORTUNITY IS AN OFFLINE PROCESSED-LOAD-PROXY OVERLAP.

Model	τ	Val. conflict	Test conflict	Safe opportunity	Safe/rec. intervals	Safe/rec. windows	kWh/day
Historical average	0.950	8.33%	0.00% (0/66)	94.6	66/66	12/12	2.20
LightGBM	0.950	9.61%	4.15% (11/265)	493.9	254/265	16/19	11.49
Primary hybrid	0.875	8.75%	0.00% (0/259)	490.1	259/259	14/14	11.40

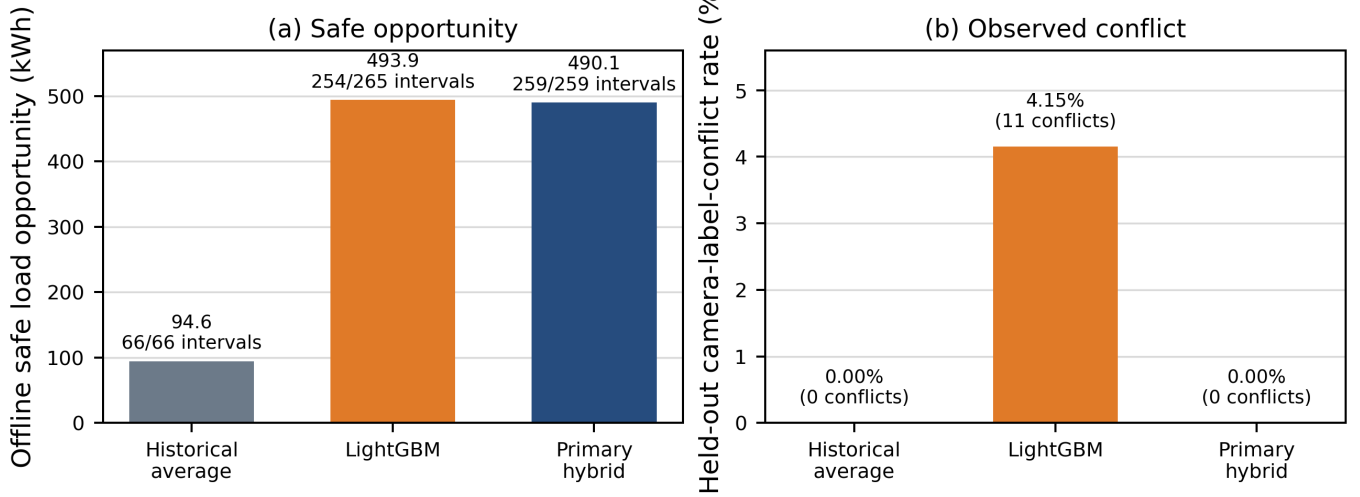


Fig. 1. Fixed held-out policy outcomes for the schedule baseline, LightGBM, and the validation-selected primary hybrid. Thresholds were selected on 39 validation daily horizons to maximize offline safe load opportunity subject to an empirical interval conflict rate of at most 10%; the plotted values come from 43 post-bin test horizons. “Safe” denotes recommended and subsequently camera-label-empty.

TABLE VI

CONDITIONAL DAILY-BLOCK UNCERTAINTY FROM 2,000 PAIRED BOOTSTRAP RESAMPLES OF 43 NON-OVERLAPPING TEST HORIZONS. DAILY-HORIZON AUPRC USES A DIFFERENT AGGREGATION FROM THE ALL-OVERLAP AUPRC IN TABLE III. ALL INTERVALS CONDITION ON SELECTED WEIGHTS AND THRESHOLDS.

Estimate type	Quantity	Point estimate	95% low	95% high
Primary model	Daily-horizon Empty AUPRC	0.8522	0.7918	0.9016
Primary model	Safe opportunity (kWh)	490.1	153.1	966.7
Paired contrast	Primary – LightGBM AUPRC	+0.0163	-0.0050	+0.0426
Paired contrast	Primary – historical AUPRC	+0.0042	-0.0304	+0.0377
Paired contrast	Primary – LightGBM opportunity (kWh)	-3.8	-208.2	+174.6
Paired contrast	Primary – historical opportunity (kWh)	+395.5	+127.0	+786.2

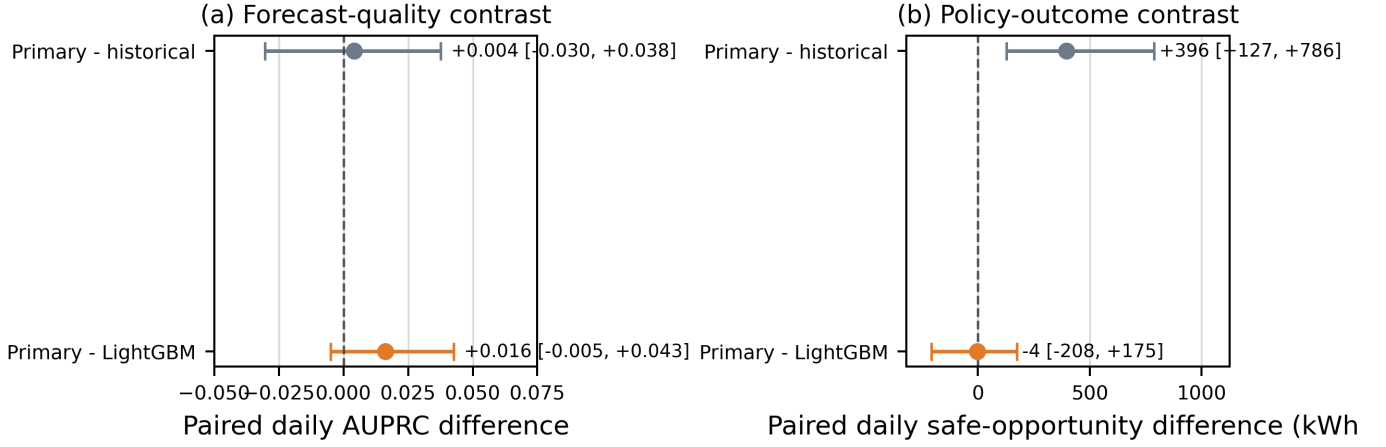


Fig. 2. Paired 95% daily-block bootstrap intervals (2,000 resamples) for the primary hybrid minus each canonical baseline. The dashed line marks zero. Intervals condition on fixed, validation-selected model weights and operating thresholds.

to estimate energy savings, flexibility capacity, carbon effects, and comfort outcomes.

B. Validity Limits

The temporal evidence supports an effective 00:15 post-bin boundary: a stored 00:00 anchor contains the completed [00:00, 00:15) input bin and targets [00:15, next 00:15). A prospective timing claim requires original source streams, observation-end timestamps, and source-side imputation lineage. The repository begins with a cleaned release whose documented upstream processing can include non-local imputation. Its post-import row-order fill establishes internal ordering from that point forward; source-causal verification remains a requirement for the prospective rerun.

The data cover selected south zones in one building and a 43-horizon test policy period. Cross-building validation is therefore the next test of generalization. Missed camera detections would label occupied intervals as Empty, so “camera-label-safe” and “conflict-free” describe agreement with the recorded label. Wi-Fi has substantial pre-fill missingness, and its forward fill uses unlimited staleness. The HVAC and lighting streams used in q are each 8.03% missing before row-order filling. Alternative sensor-availability and load-imputation patterns deserve sensitivity analysis. A solar-pattern diagnostic provides the source-timezone interpretation because the documentation leaves it unspecified.

The bootstrap conditions on selected weights and thresholds. Resampling all-zero conflict blocks yields all-zero conflict samples. Selection uncertainty is important because the nominal blend winner exceeds the runner-up by 0.000032 validation AUPRC and the validation-only perturbations in Table IV favor several nearby candidates. Deep-model construction also preceded seed reset, and the saved exports contain averaged scores in place of aligned per-seed predictions. These artifacts support post-bin validation-before-test selection and limit seed-dispersion analysis. A confirmatory evaluation would pre-

register the protocol, control model initialization, and reserve a fresh test firewall.

C. Status of Expanded Searches and Next Evidence

The repository’s decision-aware joint search and window-aware search enlarge the validation search surface, and their historical test diagnostics come from a previously inspected period. Appendix A therefore treats their point estimates as exploratory and identifies the window-aware candidate as a frozen future challenger. Decision-focused learning would additionally require downstream-objective training [19], [20].

The next empirical experiment should obtain source streams with observation-end timestamps and imputation lineage, declare a bin-end/issue-time convention, initialize seeds before model construction, lock package versions, retrain under the frozen protocol, and evaluate a later untouched period or a new building. A simulator or intervention with explicit comfort and equipment constraints can then test control and energy outcomes. This sequence addresses the operational questions examined in occupancy-aware HVAC and ventilation studies [7], [17], [18].

VII. CONCLUSION

This paper presents an offline, post-bin assessment from saved next-24-h occupancy scores to stable Empty-window recommendations. The primary hybrid combines a historical schedule, LightGBM, and a compact Transformer so that recurring, nonlinear, and sequential patterns can contribute. In one Building 59 case study, the validation-selected blend achieved 0.8514 all-overlap Empty AUPRC across 388,032 test rows. Its fixed one-hour policy identified 490.1 kWh of offline safe load opportunity with zero observed camera-label conflicts across 259 recommended test intervals. This operating point offers nearly the opportunity of LightGBM with fewer observed conflicts and much more opportunity than the conflict-free historical policy. The historical baseline’s

nearly equal AUPRC, paired intervals that cross zero, and validation perturbations identify a competitive set of blends.

The contribution is a reproducible forecast-to-policy assessment that connects discrimination, stable-window conflict, and load opportunity. A prospective evaluation with provenance-tagged source streams, corrected deep-model initialization, a locked protocol, and a later untouched period can extend this evidence to real-time operation and measured control outcomes.

APPENDIX A

EXTENDED RESULTS AND EXPLORATORY-ANALYSIS STATUS

Table VII preserves the complete canonical comparison set. The test-ranked balanced hybrid has supplementary status because its architecture was selected after inspection of the test ranking. The primary hybrid remains the nominal validation-selected canonical model.

The repository additionally contains a joint weight-threshold search over 231 convex blend weights and 37 score thresholds (8,547 validation pairs) and a window-aware search over the resulting validation surface. After the present code audit, both routines freeze validation candidates before loading their test diagnostics. Their enlarged search surface, previously inspected test period, and inherited post-bin/source-provenance limits make the decision-optimal current-test point estimates (623.5 and 650.4 kWh for two joint-search variants) and the window-aware challenger’s current-test point estimate (511.6 kWh with 2.49% interval conflict) retrospective diagnostics.

a) Status controls.: The test-ranked balanced hybrid is supplementary, the joint search is exploratory, and the window-aware candidate is a frozen future challenger. These searches select at the decision layer while retaining the fitted score models. A future evaluation can apply a frozen candidate and its declared constraints to a later untouched chronological period with control, comfort, and counterfactual energy measurements.

TABLE VII

EXTENDED CANONICAL MODEL COMPARISON ON ALL OVERLAPPING HELD-OUT TEST FORECASTS (EMPTY = 1). VALUES ARE UNCALIBRATED-SCORE DIAGNOSTICS; THE TEST-RANKED BALANCED HYBRID HAS SUPPLEMENTARY STATUS.

Model	Empty AUPRC	Empty precision	Empty recall	Empty F1
Historical average	0.8497	0.7264	0.8015	0.7621
LightGBM	0.8382	0.7610	0.7593	0.7601
Random forest	0.8329	0.7510	0.8019	0.7756
Compact Transformer encoder (legacy Transformer)	0.7621	0.7253	0.6022	0.6581
Direct linear occupancy baseline (legacy DLinear)	0.5933	0.5764	0.6657	0.6179
Seasonal-compact-encoder blend	0.8490	0.7465	0.7408	0.7436
Primary hybrid	0.8514	0.7677	0.7556	0.7616

DATA AND CODE AVAILABILITY

The cleaned Building 59 data package is available from Dryad [22]. The repository contains the canonical saved prediction and result exports and the scripts used to regenerate the manuscript figures and tables. Prospective replication additionally requires original acquisition streams, source-side imputation lineage, and timestamp provenance. A persistent archival URL must be inserted here before submission.

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